

GradeTracker

A Web-Based Grade Management System

HTML5 | CSS3 | JavaScript | PHP | MySQL

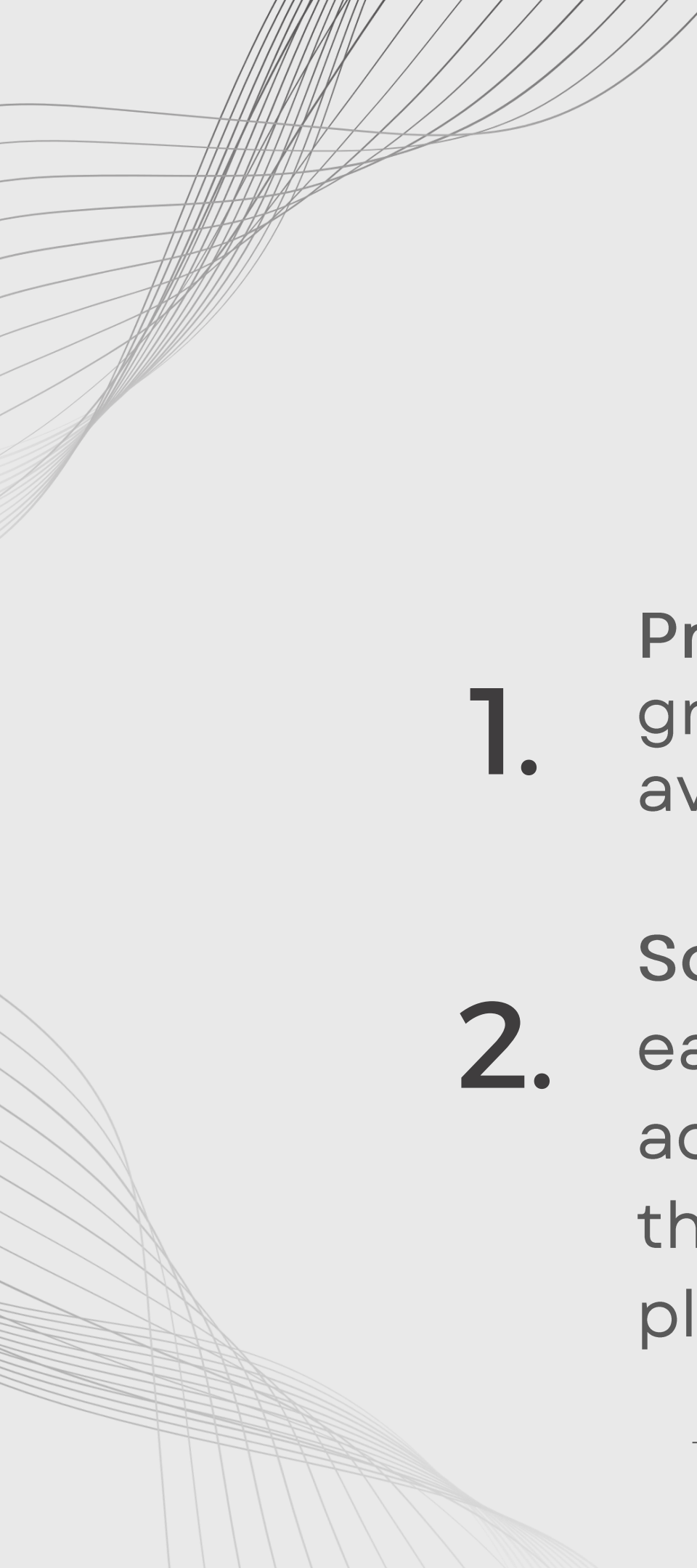


What is GradeTracker

GradeTracker is a full-stack web application that allows students to:

- Log their marks per subject and assessment
- View running averages and track academic progress
- Manage all grade records with full CRUD operations
- Visualize performance through an interactive bar chart

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Problem&Solution

1. **Problem:** Students struggle to manually track their grades across multiple subjects and calculate their averages accurately throughout the semester.
2. **Solution:** GradeTracker provides a centralized, easy-to-use platform where students can instantly add grades, view averages per subject, and monitor their overall academic performance – all in one place.

Technologies Used

Layer	Technology
Structure	HTML5 (Semantic Elements)
Styling	CSS3 (Grid, Flexbox, Media Queries)
Frontend Logic	JavaScript ES6+ (Fetch API, Array methods)
Backend	PHP 8 (PDO, Prepared Statements)
Database	MySQL
Charts	Chart.js
Server	Apache via XAMPP

Key Features

- ☒ User Registration & Login (session-based authentication)
- ☒ Add grades with subject, assessment name, mark, and total
- ☒ View all grades in a sortable, filterable table
- ☒ Edit and delete grade entries
- ☒ Subject summary sidebar with average and letter grade
- ☒ Overall GPA / average display
- ☒ Performance bar chart (green = pass, red = fail)
- ☒ Fully responsive on mobile, tablet, and desktop

HTML5 Structure

The application uses proper semantic HTML5 elements throughout:

- <header> – wraps the navigation bar
- <nav> – site navigation with anchor links
- <main> – main content area
- <section> – Add Grade form and Grade Records
- <aside> – Subject Summary sidebar panel
- <table> with <caption>, <thead>, <tbody> – grade records table
- <dialog> – accessible Edit Grade modal
- <footer> – page footer
- All form inputs have associated <label> elements

CSS3 Layout & Design

CSS Grid – Dashboard Layout:

CSS

```
.dashboard {  
  display: grid;  
  grid-template-columns: 1fr 300px; }
```

Main grade table on the left, summary sidebar on the right.

CSS Flexbox – Summary Cards: Each subject card uses Flexbox to display subject name, average percentage, and letter grade in a row.

Color Coding:

-  Green (#4caf50) – grades above 50% (passing)
-  Red (#d43f3f) – grades below 50% (failing)

Responsive Design:

- Tablet (max 900px): sidebar stacks below the table
- Mobile (max 600px): navbar stacks vertically, table font shrinks

JavaScript

Fetch API: All data is loaded and submitted asynchronously – no page reloads.

Javascript

```
const res = await fetch("backend/getGrades.php"); const grades = await res.json();
```

Array Methods:

- Array.map() – render table rows dynamically
- Array.sort() – sort grades by any column on click
- Array.filter() – filter grades by selected subject
- Array.reduce() – calculate overall average

DOM Manipulation: Table rows, summary cards, and the GPA block are all rendered dynamically by JavaScript based on data from the backend.

PHP Backend & Security

PDO with Prepared Statements: All database queries use PDO prepared statements to prevent SQL injection.

Php

```
$stmt = $conn->prepare("SELECT * FROM grades WHERE user_id = ?"); $stmt->execute([$SESSION["user_id"]]);
```

Input Validation:

- Mark and total are validated as numeric
- Mark must be between 0 and total
- Empty fields are rejected with clear error messages

Session Authentication:

- Every PHP endpoint checks `$_SESSION["user_id"]`
- Unauthorized requests return HTTP 401
- Passwords are hashed using `password_hash()`
- Session is regenerated on login to prevent session fixation

JSON Responses: All endpoints return structured JSON – not HTML – for clean frontend integration.

Database Schema

Table: users

Column	Type	Description
id	INT (PK)	Auto increment
username	VARCHAR(100)	Unique username
password	VARCHAR(255)	Hashed password
created_at	TIMESTAMP	Registration date

Table: grades

Column	Type	Description
id	INT (PK)	Auto increment
user_id	INT (FK)	References users.id
subject	VARCHAR(100)	Subject name
assessment	VARCHAR(100)	Assessment name
mark	FLOAT	Student's mark
total	FLOAT	Total possible mark
created_at	TIMESTAMP	Entry date

Relationship: One user → Many grades (CASCADE DELETE)



Responsive Design

the application is tested and works across 3 screen sizes:

Desktop (1440px): Grid layout – grade table on the left, summary sidebar on the right.

Tablet (768px): Sidebar stacks below the grade table. Full functionality maintained.

Mobile (375px): Navbar collapses vertically. Table font size reduces. Cards stack vertically.

All breakpoints handled with CSS @media queries.



Technical Challenge & Solution

Challenge: When switching from HTML form submissions to the Fetch API, grade additions stopped working. The PHP endpoints were returning Unauthorized even after a successful login.

Root Cause: Opening index.html directly as a static file bypassed the PHP session entirely – no session cookie was being created.

Solution: Ensured the user always enters the app through login.php, which creates the session properly. All subsequent Fetch requests automatically include the session cookie, so every PHP endpoint correctly identifies the logged-in user.

Lesson Learned: Understanding how PHP sessions interact with static files and the browser's cookie handling is essential for building secure full-stack applications.

Live Demo

Demo flow:

- 1.Register a new account
- 2.Login
- 3.Add 3-4 grades across different subjects
- 4.Show the grade table – sort by column, filter by subject
- 5.Edit a grade
- 6.Delete a grade
- 7.Show Subject Summary sidebar and Overall Average
- 8.Show the Performance Chart

<https://drive.google.com/file/d/18eXJS0kF0e4EgYII9YhFCcBd0gUsx5xi/view?usp=drivesdk>

CONCLUSION

What was built: A complete, secure, and responsive full-stack grade management application using HTML5, CSS3, JavaScript, PHP, and MySQL.

What was learned:

- Building RESTful PHP endpoints that return JSON
- Using the Fetch API for seamless frontend-backend communication
- Managing user sessions securely across a full-stack application
- Applying CSS Grid and Flexbox for responsive layouts

<https://gradetracker.kesug.com/> (live link)



THANK YOU